MCV4U1 – V2 **Name:** Solutions **Alpha #:**____

Chang, Moshtagh, Vincent

Unit 4 Assessment of Learning - Application of Derivatives

Knowledge & Understanding	Application	Thinking	Comm.
/ 11	/10	/ 5	/2

Instructions: Please show all your work for full marks. You may only use methods taught in MCV4U for full marks. Express your final answers in **exact** form, unless otherwise indicated. The use of cellphones, audio-or video-recording devices, digital music players or email or text-messaging devices during the assessment is prohibited.

KNOWLEDGE & UNDERSTANDING - [11 Marks]

1. A position function of an object is given by $s(t) = -\frac{1}{3}(t-2)^3 + 4t, t \geq 0$, where t is in seconds and $s(t)$ is in metres.

a. Determine the **exact** initial velocity of the object. **[1 Mark]**

$v(t) = -(t-2)^2 + 4$

$v(0) = 0 \text{ m/s}$

b. Determine the acceleration of the object when the velocity is $4\, m/s$. **[3 Marks]**

$-(t-2)^2 + 4 = 4$

$(t-2)^2 = 0$

$t = 2$

$a(t) = -2(t-2)$

$a(2) = 0 \text{ m/s}^2$

c. Is the object speeding up or slowing down when $t = 6$ seconds? **[3 Marks]**

$v(t)a(t)$

$= [-(t-2)^2 + 4][-2(t-2)]$

$= [-(4)^2 + 4][-8]$

$= (-12)(-8)$

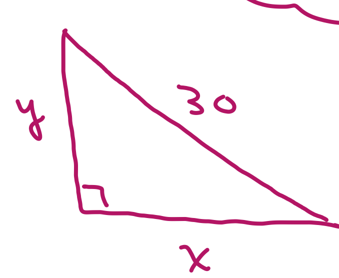
$= 96 > 0$

$\therefore$ object is speeding up at $t = 6$ s.

2. Determine the **exact** maximum perimeter of a right triangle with hypotenuse 30 cm. **[4 Marks]**

$x^2 + y^2 = 900$

$y = \sqrt{900 - x^2}$



$P = x + y + 30$

$P = x + \sqrt{900 - x^2} + 30$

$P' = 1 - \dfrac{x}{\sqrt{900 - x^2}}$

$P' = 0$

$1 - \dfrac{x}{\sqrt{900 - x^2}} = 0$

$900 - x^2 = x^2$

$2x^2 = 900$

$x = \sqrt{450}$; $y = \sqrt{450}$

$\therefore$ Max Perimeter $= 2\sqrt{450} + 30$ cm

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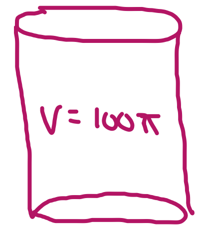
APPLICATION - [10 Marks]

1. An **open top** steel cylindrical container is to be constructed to hold $100\pi\ m^3$ of water. The cost of constructing the sides of the container is $\$6/m^2$ and the cost of constructing the bottom of the container is $\$8/m^2$. Determine the dimensions of the container that will **minimize** the cost of construction. Round your answers to 2 decimal places. **[5 Marks]**

$V = \pi r^2 h$

$100\pi = \pi r^2 h$

$* \ h = \dfrac{100}{r^2}$



$SA = \underbrace{\$8\pi r^2}_{\text{bottom}} + \underbrace{\$6\cdot(2\pi r h)}_{\text{side}}$

$SA = 8\pi r^2 + 12\pi r \cdot \dfrac{100}{r^2}$

$SA = 8\pi r^2 + \dfrac{1200\pi}{r}$

$SA' = 16\pi r - \dfrac{1200\pi}{r^2}$

$SA' = 0$

$16\pi r^3 - 1200\pi = 0$

$r^3 = \dfrac{1200}{16}$

$r^3 = 75$

$r = \sqrt[3]{75}$

$r \doteq 4.22\,m$

$h \doteq 5.62\,m$

2. One leg of a right triangle is on the positive x-axis and the endpoints of its hypotenuse are on the graph of $f(x) = -2x^2 + 24x$. Determine the **exact** maximum area of the triangle. **[5 Marks]**

$A = \frac{1}{2} b \cdot h$

$A = \frac{1}{2} x\left(-2x^2 + 24x\right)$

$A = \frac{1}{2}\left(-2x^3 + 24x^2\right)$

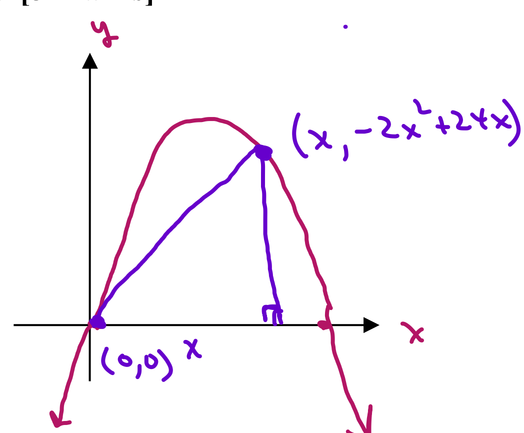
$A' = \frac{1}{2}\left(-6x^2 + 48x\right)$

$A' = 0 \Rightarrow -6x^2 + 48x = 0$

$-6x(x-8) = 0$

$x = 0, 8$

reject



$A = \frac{1}{2}(8)(-128 + 192)$

$A = 256 \text{ units}^2$

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THINKING - [5 Marks]

1. Marvin is building a window in the shape of an equilateral triangle whose sides each measure 4 metres. He wants to inscribe a rectangular piece of stained glass in the triangle so that two of the vertices of the rectangle lie on one of the sides of the triangle. Determine the **exact** dimensions of the rectangle of maximum area that can be inscribed in the given triangle. **[5 Marks]**

$A = 2xy$

$A = 2x\left(\dfrac{\sqrt{12}(2-x)}{2}\right)$

$A = 2\sqrt{12}x - \sqrt{12}x^2$

$A' = 2\sqrt{12} - 2\sqrt{12}x$

$A' = 0$

$2\sqrt{12} - 2\sqrt{12}x = 0$

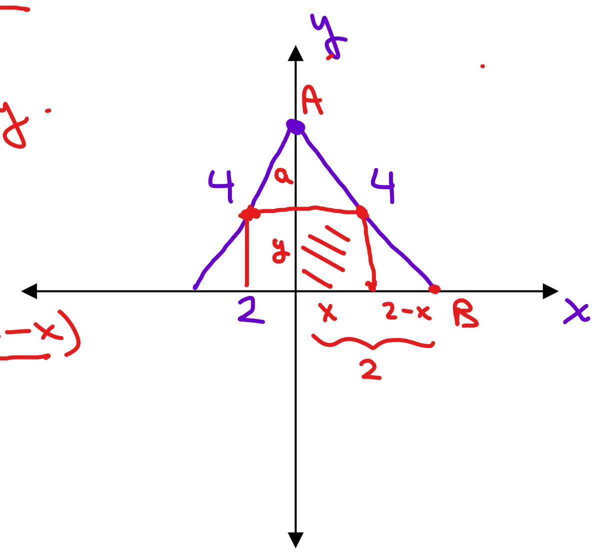
$x = 1$

$a + y = \sqrt{12}$

$a = \sqrt{12} - y$.

$\dfrac{\sqrt{12}}{y} = \dfrac{2}{2-x}$

$y = \dfrac{\sqrt{12}(2-x)}{2}$



$y = \dfrac{\sqrt{12}(2-1)}{2}$

$y = \sqrt{3}$

$\therefore$ Dimensions are

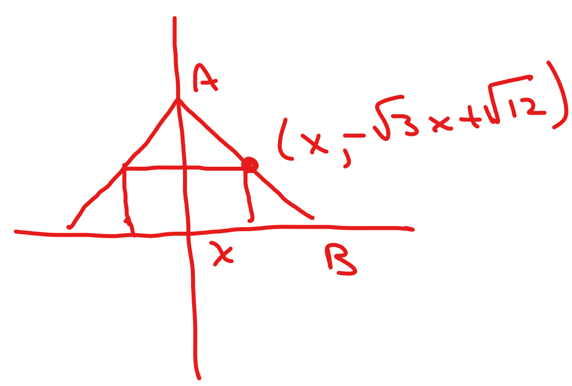
2 units $\times \sqrt{3}$ units.

Solution #2

Get Equation of AB.

$m = -\dfrac{\sqrt{12}}{2} = -\sqrt{3}$ $\quad$ $y\text{-int} = \sqrt{12}$

$\therefore y = -\sqrt{3}x + \sqrt{12}$



$A = 2xy$

$A = 2x(-\sqrt{3}x + \sqrt{12})$

$A = -2\sqrt{3}x^2 + 2\sqrt{12}x$

$A' = -4\sqrt{3}x + 2\sqrt{12}$

$A' = 0$

$-4\sqrt{3}x + 2\sqrt{12} = 0$

$x = \dfrac{2\sqrt{12}}{4\sqrt{3}}$

$x = \dfrac{4\sqrt{3}}{4\sqrt{3}}$

$x = 1$

$y = -\sqrt{3}(1) + \sqrt{12}$

$y = \sqrt{3}$

Dimensions are.

2 units $\times \sqrt{3}$ units

*** 2 Marks will be awarded in the Communication Category for proper mathematical form. ***